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MIL-STD-2189(SH)
SECTION 305-1
7 April 1988

MILITARY STANDARD

DESIGN METHODS FOR NAVAL SHIPBOARD SYSTEMS

SECTION 305-1

DESIGNATION AND MARKING OF ELECTRIC SYSTEM

AMSC N/A
FSC 1990
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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362-5101

Design Methods for Naval Shipboard Systems,
Designation and Marking of Electric System

1. This Military Standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

Purpose. The purpose of this standard is to set forth design methods for shipboard components or systems as designated in the title.

Background. This section was first developed as a Navy design data sheet, DDS 305-1, but is being published as a military standard to increase its availability throughout the Department of Defense.

Concept. This section provides examples of how shipboard electric systems and components should be designated, numbered, and tagged.

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1. GENERAL AND SCOPE

1.1 General. The procedures established by MIL-STD-2189(SH) are applicable. This section and the basic standard are to be considered as an integral single document.

1.2 Scope. This section gives examples of designation and numbering of electric systems on various types of ships, and provides acceptable methods of attaching tags to cables.

2. REFERENCED DOCUMENTS

Not applicable.

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENT

4.1 Systems applicable. Power, lighting, degaussing, and other electric equipment circuits, including cables, shall be clearly labeled so that they can be quickly identified and readily distinguished from one another.

5. DETAILED REQUIREMENTS

5.1 Working examples. Examples are shown on figures 1 through 6. The examples given are based on portions of typical ship systems and will need further development for specific applications. Each system of new design will differ in some detail. The examples are intended to anticipate as many such differences as possible (see 6.3).

6. NOTES

6.1 Intended use. This standard is intended for use in implementing the requirements for designating, numbering, and tagging electric systems and components on Navy ships.

6.2 Information. Rules for numbering and otherwise designating electric systems and their components for Navy ships are set by Commander, Naval Sea Systems Command, and are changed from time to time to meet the changing needs of the fleet.

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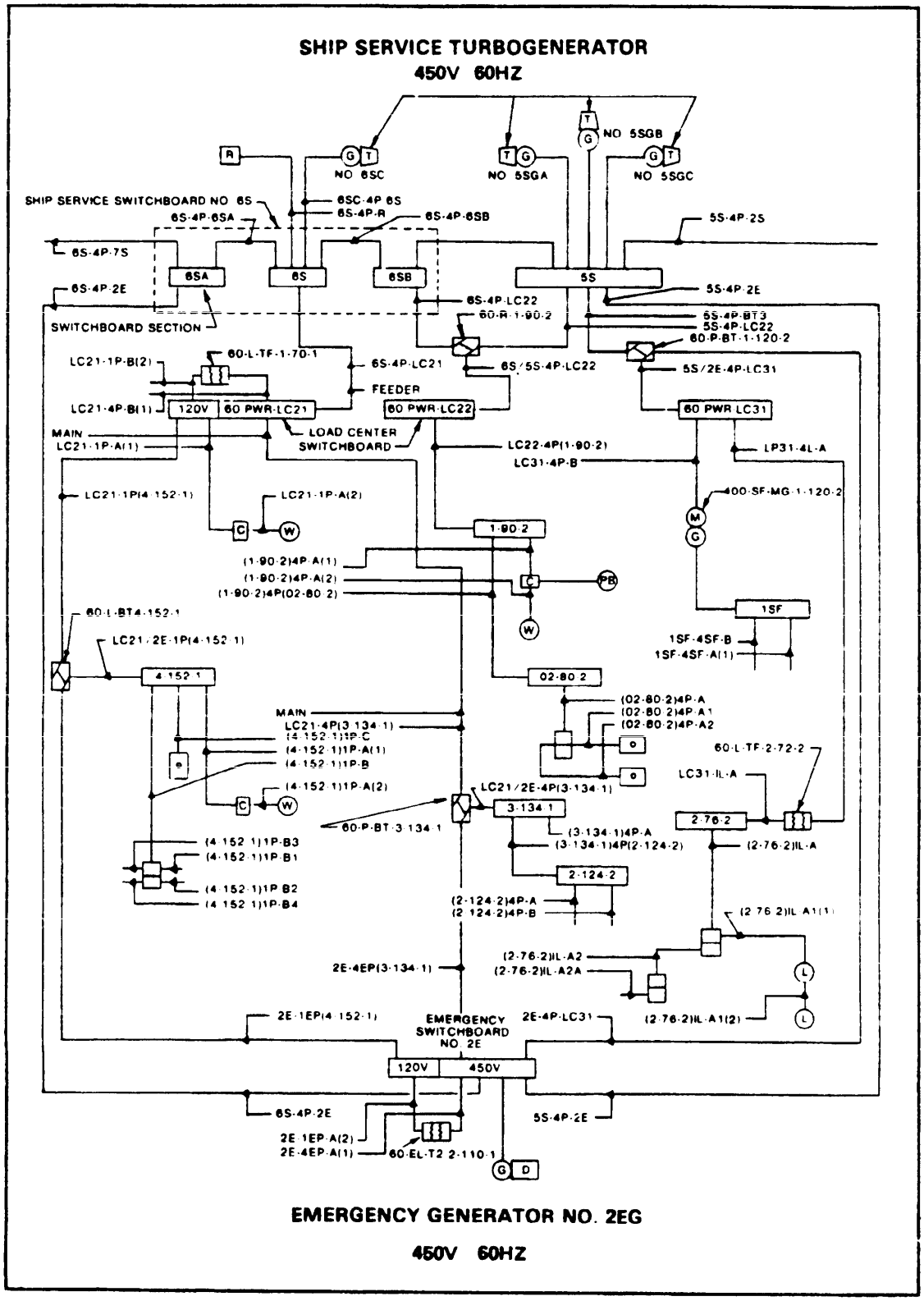
6.3 Working examples. In case of questions involving apparent departure from the specified principles, approval of the contracting officer should be obtained.

6.4 Subject term (key word) listing.

Electric cables
Electric conductors
Electric connectors
Identification tags
Labels

Preparing activity:
Navy - SH
(Project 1990-N061)

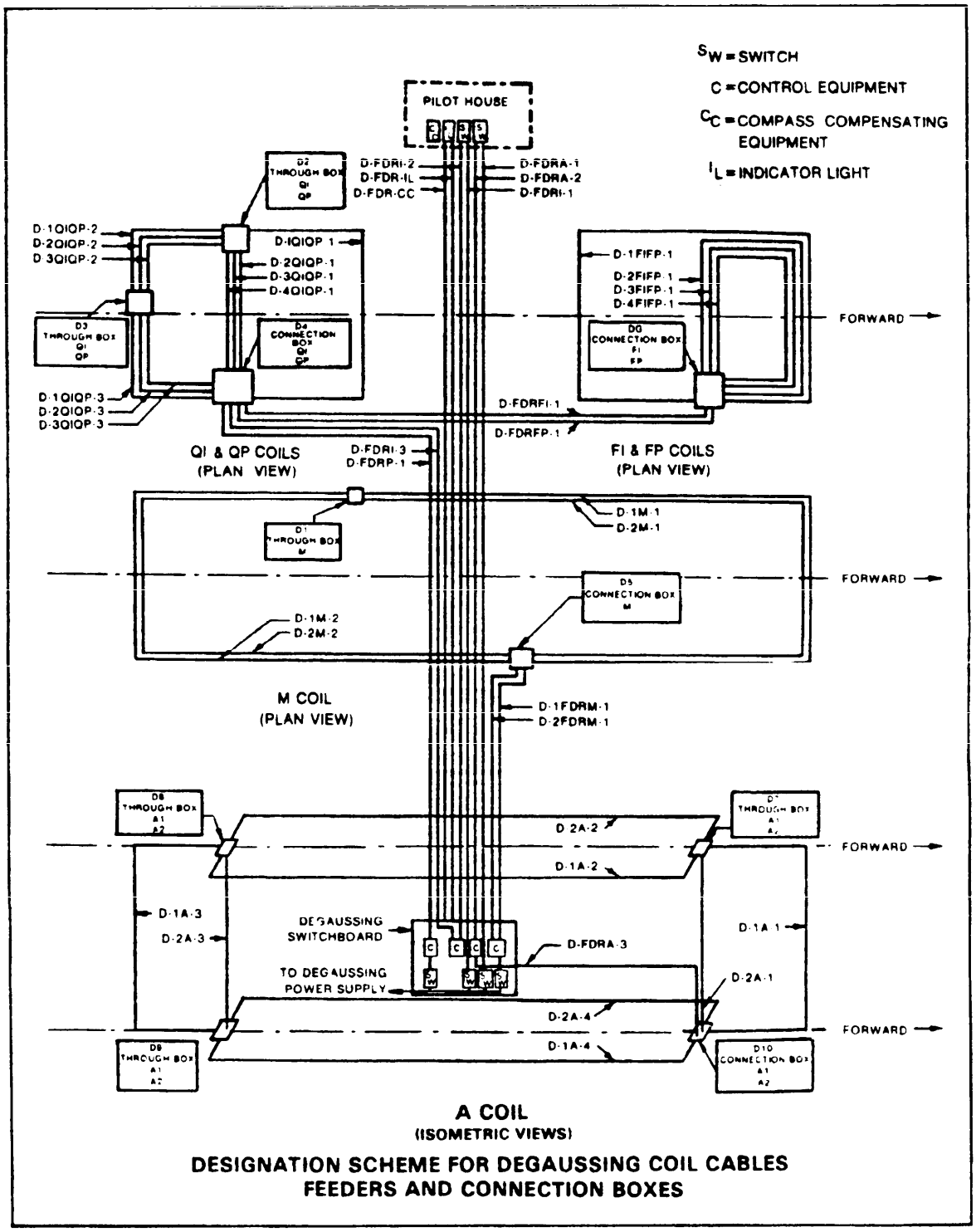
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FIGURE 1. Typical designation of a power and lighting system for a surface ship.

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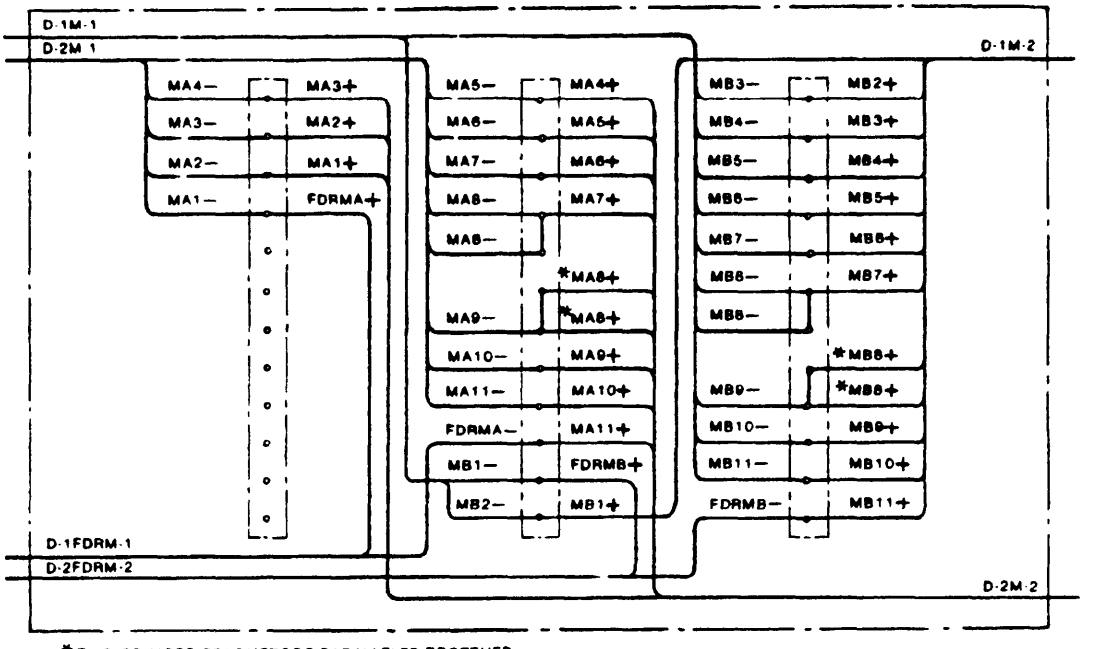


DESIGNATION SCHEME FOR DEGAUSSING COIL CABLES FEEDERS AND CONNECTION BOXES

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FIGURE 2. Designation scheme for degaussing coil cables, feeders, and connection boxes.

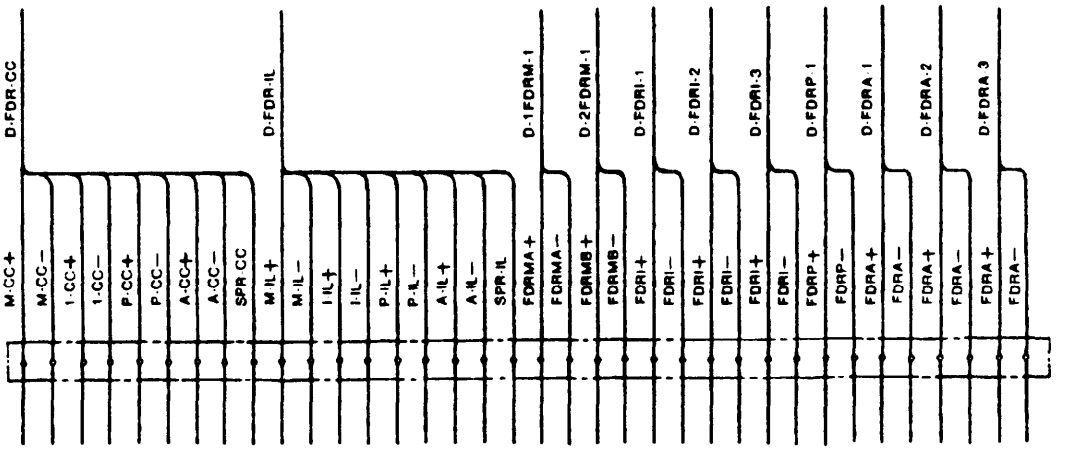
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*TWO OR MORE CONDUCTORS PARALLELED TOGETHER WITHIN A PARTICULAR SERIES OR PARALLEL CIRCUIT RECEIVE THE SAME CONDUCTOR DESIGNATION AS AN EQUIVALENT SINGLE CONDUCTOR WOULD RECEIVE

12 CONDUCTOR CABLE USED FOR M COIL

TYPICAL DEGAUSSING SWITCHBOARD CONNECTION STRIP



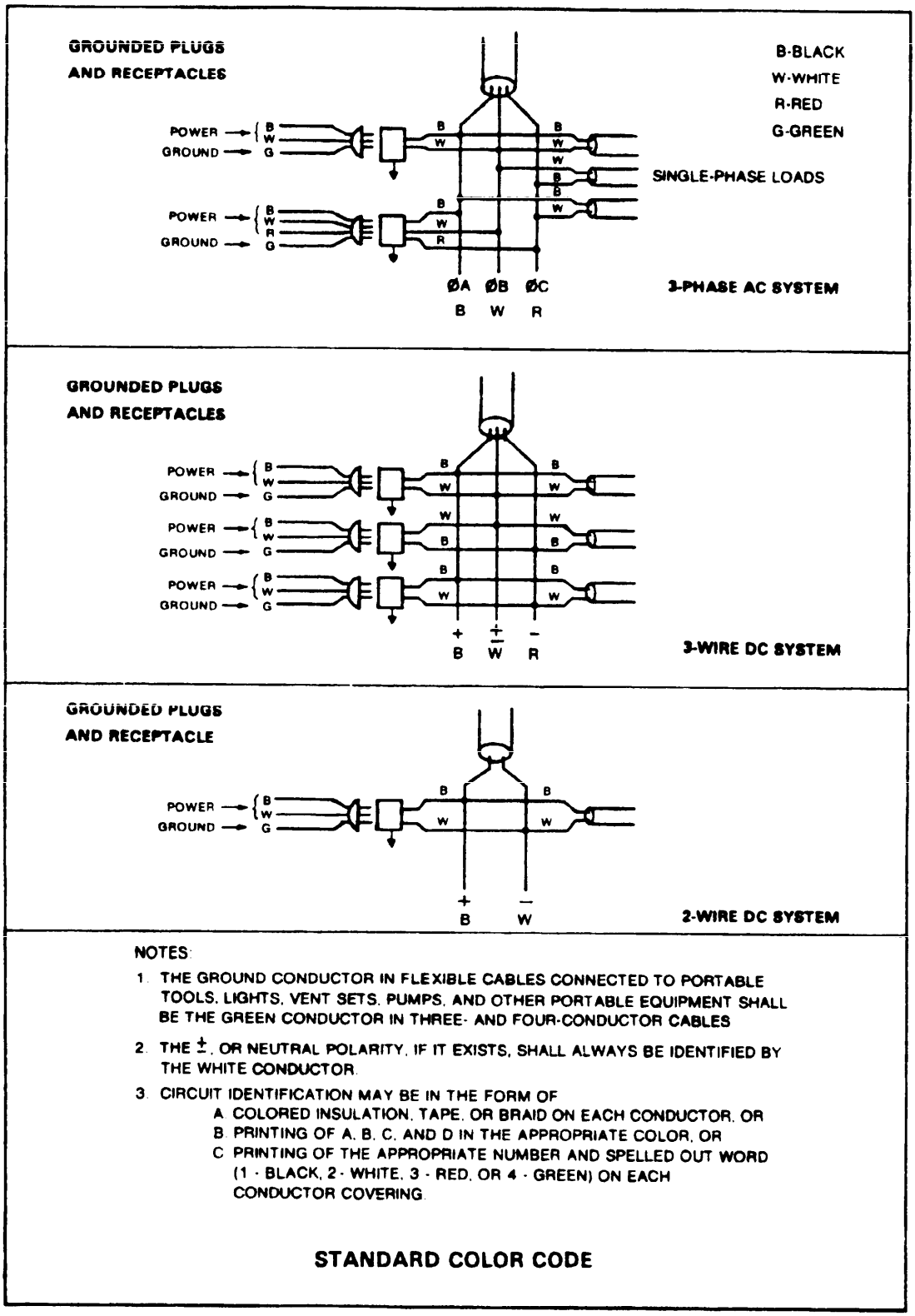
TYPICAL SINGLE LOOP CONNECTION BOX

D5 CONNECTION BOX (M COIL) STARBOARD SIDE, VIEWER LOOKING TO STARBOARD

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FIGURE 3. Typical single loop connection box D5 (M coil) starboard side, viewer looking to starboard.

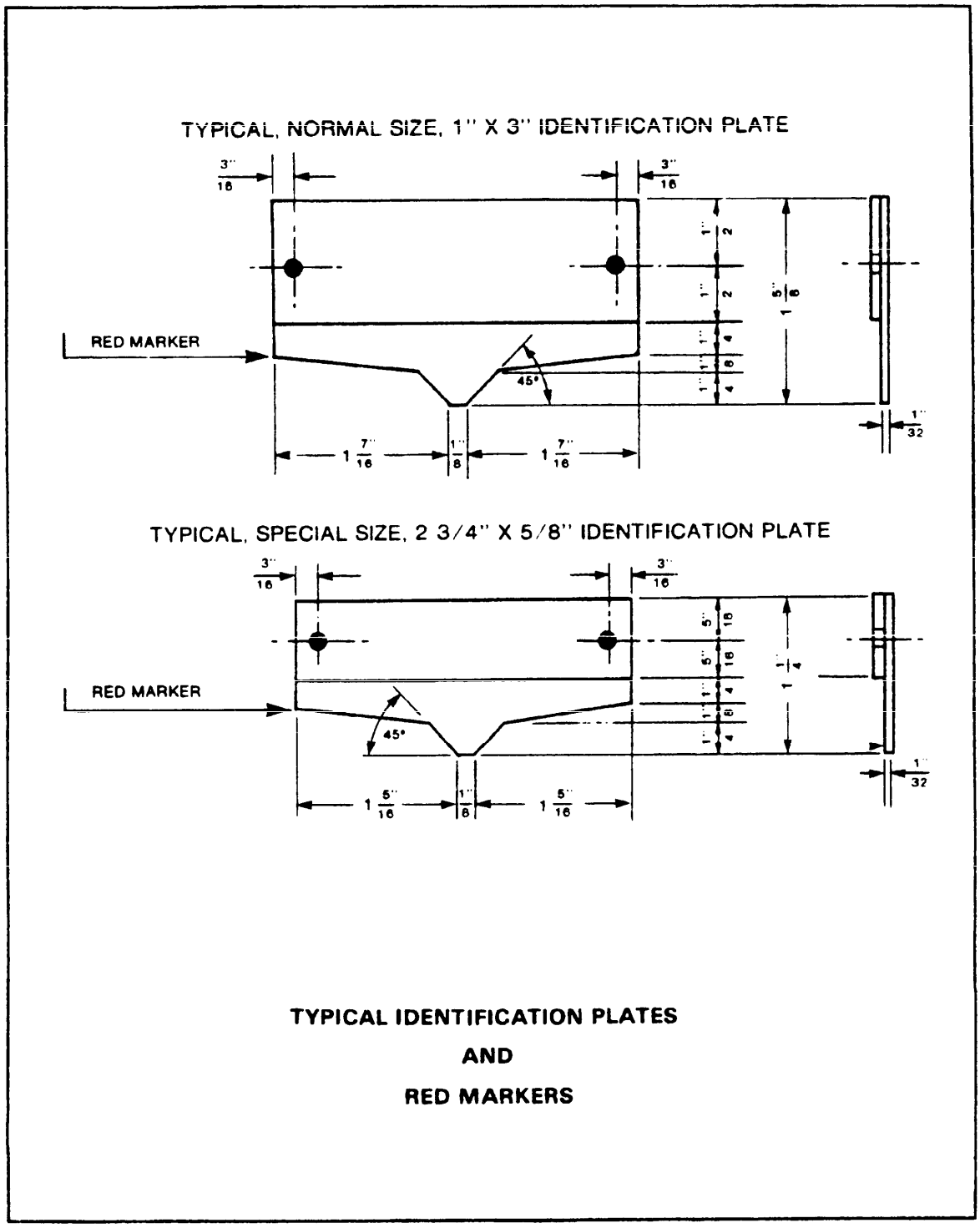
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FIGURE 4. Standard color code.

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TYPICAL IDENTIFICATION PLATES

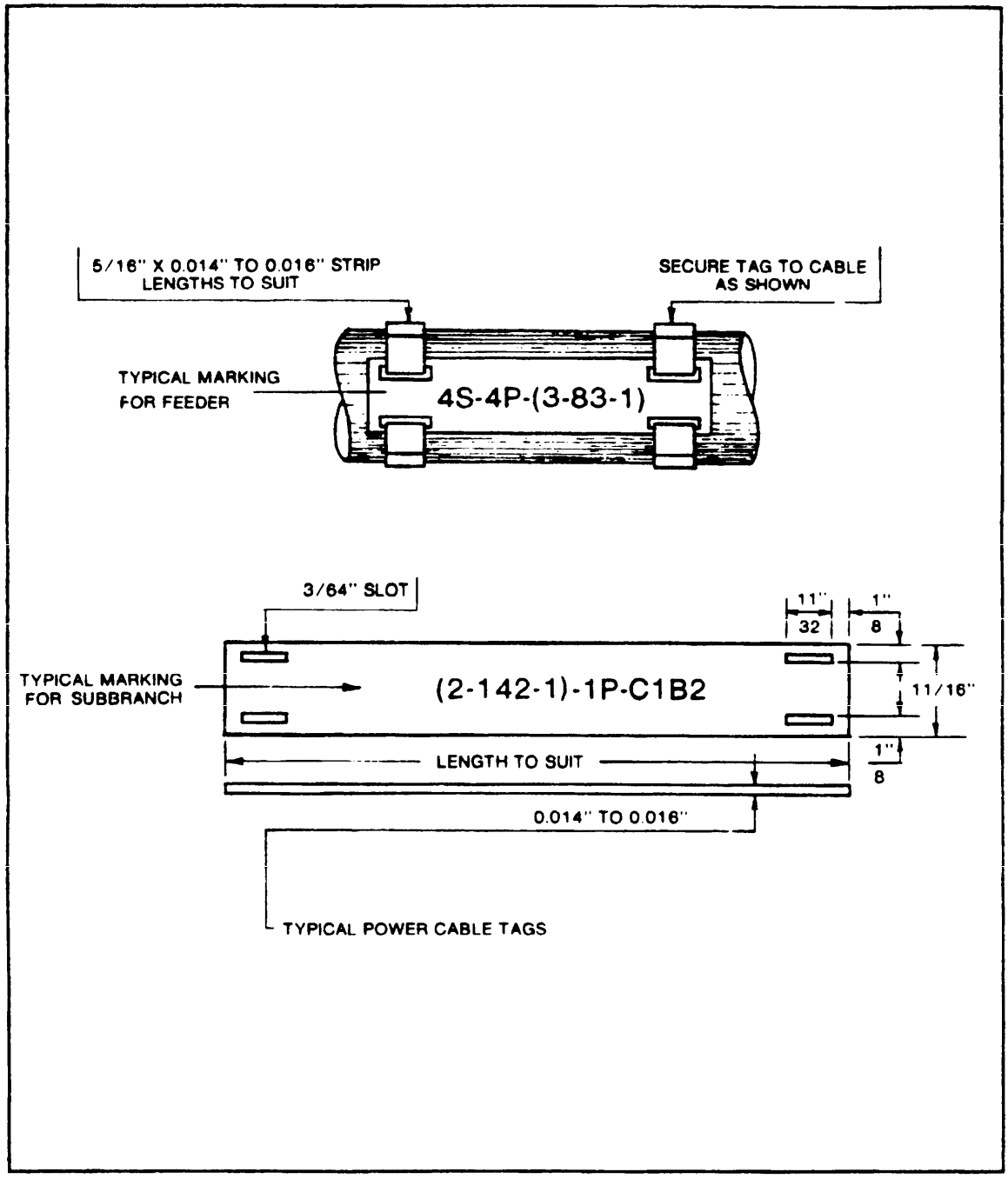
AND

RED MARKERS

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FIGURE 5. Typical identification plates and red markers.

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FIGURE 6. Typical cable tags.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-2189(SH) SECTION 305-1	Designation and Marking of Electric System

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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